

HAT-P-54b

R-1102

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_241228 · created 2026-07-30T19:40:13+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460672.9366 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.054
Transit depth (Rp/Rs)^2	3.39 +/- 1.98 [%]
Orbital Inclination (inc)	89.74 +/- 2.08
Airmass coefficient 1 (a1)	0.911 +/- 0.021
Airmass coefficient 2 (a2)	0.06 +/- 0.042
Scatter in the residuals of the lightcurve fit is	2.31 %
Best Comparison Star	#3 - [607, 412]
Optimal Aperture	2.3
Optimal Annulus	14.55
Transit Duration (day)	0.0348 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.054 vs 0.1572 (deviation 0.5σ)
Duration fit vs published	0.0348 d vs 0.075 d (deviation 7.44σ)
Mid-time O–C	1.5 min
Depth SNR	3.4
Residual scatter	2.31 %

Light curve

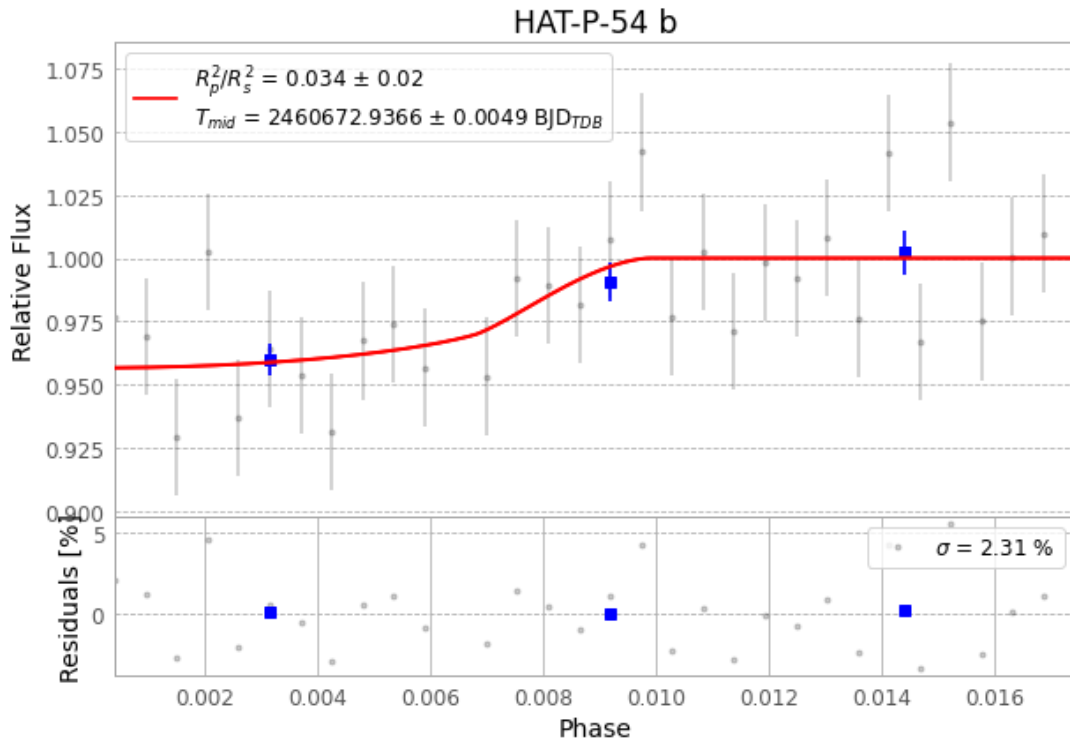


Figure 1 — FinalLightCurve_HAT-P-54 b_2024-12-28.png (SHA-256 03fadb1a94ea953f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 315], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 4850cb93cf25d98e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

32 FITS frames (21 MB), HATP-54241228102115.FITS → HATP-54241228115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-241228.log (708d89787bc2...), FinalLightCurve_HAT-P-54 b_2024-12-28.png (03fadb1a94ea...), FinalLightCurve_HAT-P-54 b_2024-12-28.csv (0ca6de9f6e57...)

Compute resources

Wall time 90.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 19:40Z.